

Selected Advanced Topics in Mathematical Analysis

A Rigorous Treatment of Selected Chapters
from the Mathematical Analysis Curriculum

Independent Research Monograph

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This work was prepared independently as a deep exploration of core topics
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Abstract

This monograph offers a rigorous, self-contained development of selected advanced topics from the Mathematical Analysis curriculum. The exposition begins with the theory of uniform convergence and its consequences for the interchange of limits, integration and differentiation, and proceeds through a careful treatment of the Riemann integral (including improper integrals and integrals depending on a parameter), the differential calculus of functions of several variables, multiple integrals, Fourier series, the elementary theory of metric spaces and compactness in Euclidean space, and the Picard–Lindelöf existence and uniqueness theorem for ordinary differential equations.

All principal results are proved in full from first principles. The arguments remain within the classical framework of the real line and Euclidean space, with metric-space language introduced only when it clarifies the underlying structure. The text is written for a reader who has already mastered a first rigorous course in single-variable analysis.

The author has aimed at a style that is precise yet readable, preferring direct elementary reasoning whenever it is available, while still pointing toward the more abstract viewpoints that become natural at a later stage of study.

Keywords: uniform convergence, Riemann integral, multivariable calculus, multiple integrals, Fourier series, metric spaces, Heine-Borel theorem, Picard–Lindelöf theorem, ordinary differential equations.

Preface

These notes grew out of a desire to understand, at a deeper level than is ordinarily required in the second-year course, a number of classical theorems that form the core of real analysis. Uniform convergence, the finer properties of the Riemann integral, the differential calculus in several variables, the foundations of multiple integration, the elementary theory of Fourier series, the geometry of compact sets in Euclidean space, and the existence theory for ordinary differential equations all appear in every serious treatment of the subject, yet they are often presented in a somewhat compressed form. My aim has been to develop each of these topics carefully and completely, with full proofs and with attention to the natural connections between them.

I have preferred direct, elementary arguments whenever they are available, while still indicating the more abstract perspectives that become natural later. Occasional remarks of a personal character appear in the text; they simply record the way I myself came to see certain points and should not be read as authoritative claims.

The work was written while I was a second-year undergraduate student of Mathematics at Baku State University. The curriculum of the Mathematical Analysis course provided both the original motivation and the guiding framework. I am grateful to that tradition.

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Chapter 1

Uniform Convergence of Sequences of Functions

1.1 Pointwise versus uniform convergence

Let $I \subset \mathbb{R}$ be an interval (bounded or unbounded) and let $\{f_n\}$ be a sequence of real-valued functions defined on I . We say that the sequence *converges pointwise* on I to a function $f : I \rightarrow \mathbb{R}$ if, for every $x \in I$,

$$\lim_{n \rightarrow \infty} f_n(x) = f(x).$$

In other words, for each fixed x the numerical sequence $\{f_n(x)\}$ converges to the number $f(x)$.

Pointwise convergence is a natural notion, yet it is often too weak to preserve important analytic properties. The classical counter-example is well known: the sequence

$$f_n(x) = x^n, \quad x \in [0, 1],$$

converges pointwise to the discontinuous function

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1, \\ 1 & \text{if } x = 1. \end{cases}$$

Thus the pointwise limit of continuous functions need not be continuous.

A stronger mode of convergence is required. We say that $\{f_n\}$ *converges uniformly* on I to f if

$$\lim_{n \rightarrow \infty} \sup_{x \in I} |f_n(x) - f(x)| = 0.$$

Equivalently, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies |f_n(x) - f(x)| < \varepsilon \quad \text{for all } x \in I.$$

The essential point is that the same N works simultaneously for every point of the interval.

Remark 1.1.1. Uniform convergence can be rephrased in the language of the supremum norm. If we set

$$\|g\|_I := \sup_{x \in I} |g(x)|$$

(allowing the value $+\infty$), then uniform convergence of f_n to f on I is precisely the statement that

$$\|f_n - f\|_I \rightarrow 0.$$

Proposition 1.1.2 (Uniform limit of continuous functions). *Suppose each f_n is continuous on an interval I and $f_n \rightarrow f$ uniformly on I . Then f is continuous on I .*

Proof. Fix an arbitrary point $a \in I$ and an arbitrary $\varepsilon > 0$. Choose N so that

$$\|f_N - f\|_I < \frac{\varepsilon}{3}.$$

Since f_N is continuous at a , there exists $\delta > 0$ such that

$$|x - a| < \delta, x \in I \implies |f_N(x) - f_N(a)| < \frac{\varepsilon}{3}.$$

Then, for all $x \in I$ with $|x - a| < \delta$,

$$\begin{aligned} |f(x) - f(a)| &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(a)| + |f_N(a) - f(a)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Thus f is continuous at a . Since a was arbitrary, f is continuous on I . $\square$

The same argument shows that uniform convergence preserves uniform continuity on an interval.

1.2 The Cauchy criterion for uniform convergence

A sequence of functions $\{f_n\}$ is said to be *uniformly Cauchy* on I if for every $\varepsilon > 0$ there exists N such that

$$m, n \geq N \implies \|f_m - f_n\|_I < \varepsilon.$$

Theorem 1.2.1 (Cauchy criterion). *A sequence $\{f_n\}$ of real-valued functions on I converges uniformly on I if and only if it is uniformly Cauchy on I .*

Proof. Necessity is immediate from the triangle inequality for the supremum norm.

For sufficiency, assume $\{f_n\}$ is uniformly Cauchy. Then for each fixed $x \in I$ the numerical sequence $\{f_n(x)\}$ is Cauchy and therefore converges to some real number, which we denote by $f(x)$. It remains to show that the convergence $f_n \rightarrow f$ is uniform.

Let $\varepsilon > 0$ and choose N so that $\|f_m - f_n\|_I < \varepsilon$ whenever $m, n \geq N$. Fix $n \geq N$ and let $m \rightarrow \infty$. For every $x \in I$ we obtain

$$|f(x) - f_n(x)| = \lim_{m \rightarrow \infty} |f_m(x) - f_n(x)| \leq \varepsilon.$$

Hence $\|f - f_n\|_I \leq \varepsilon$ for all $n \geq N$, which is precisely uniform convergence. $\square$

Remark 1.2.2. The completeness of $\mathbb{R}$ is used in an essential way: it guarantees that a uniformly Cauchy sequence converges pointwise, after which the uniform estimate passes to the limit.

1.3 The Weierstrass M -test

The most useful practical criterion for uniform convergence of series is the following.

Theorem 1.3.1 (Weierstrass M -test). *Let $\{f_n\}$ be a sequence of functions on a set $E \subset \mathbb{R}$ and suppose there exists a sequence of non-negative constants $\{M_n\}$ such that*

$$|f_n(x)| \leq M_n \quad \text{for all } x \in E \text{ and all } n,$$

and such that the numerical series $\sum M_n$ converges. Then the series $\sum f_n$ converges absolutely and uniformly on E .

Proof. Absolute convergence at each point follows at once by the comparison test. To obtain uniform convergence, observe that the partial sums

$$s_n(x) = \sum_{k=1}^n f_k(x)$$

satisfy, for $m > n$,

$$|s_m(x) - s_n(x)| \leq \sum_{k=n+1}^m |f_k(x)| \leq \sum_{k=n+1}^m M_k.$$

The right-hand side tends to zero as $n \rightarrow \infty$ independently of x , by the Cauchy criterion for the convergent series $\sum M_n$. Thus $\{s_n\}$ is uniformly Cauchy and therefore converges uniformly. $\square$

Example 1.3.2. The series

$$\sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}$$

converges uniformly on the whole real line, since

$$\left| \frac{\sin(nx)}{n^2} \right| \leq \frac{1}{n^2}$$

and $\sum n^{-2}$ converges.

1.4 Interchange of limit and integral

One of the most important applications of uniform convergence is the justification of passage to the limit under the integral sign.

Theorem 1.4.1. *Let $\{f_n\}$ be a sequence of Riemann-integrable functions on a compact interval $[a, b]$ that converges uniformly to a function f . Then f is Riemann-integrable on $[a, b]$ and*

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx.$$

Proof. Uniform convergence of continuous functions (or, more generally, of Riemann-integrable functions) implies that the limit f is Riemann-integrable; we recall the standard argument via the Cauchy criterion for the upper and lower sums, or via the fact that the set of discontinuities of f

is contained in the union of the sets of discontinuities of the f_n together with a set of measure zero arising from the uniform limit.

Once integrability of f is granted,

$$\left| \int_a^b f_n - \int_a^b f \right| = \left| \int_a^b (f_n - f) \right| \leq \int_a^b |f_n - f| \leq (b-a) \|f_n - f\|_{[a,b]}.$$

The right-hand side tends to zero by uniform convergence. $\square$

Remark 1.4.2. The hypothesis of uniformity cannot be omitted. The classical sequence of “moving bumps”

$$f_n(x) = n \cdot \mathbf{1}_{[0,1/n]}(x)$$

on $[0, 1]$ converges pointwise to zero, yet

$$\int_0^1 f_n = 1 \not\rightarrow 0.$$

1.5 Term-by-term differentiation

A parallel result holds for differentiation, but an additional hypothesis is needed.

Theorem 1.5.1. *Suppose that each f_n is continuously differentiable on an open interval I , that the series $\sum f_n(x_0)$ converges for at least one point $x_0 \in I$, and that the series of derivatives $\sum f'_n$ converges uniformly on every compact subinterval of I . Then $\sum f_n$ converges uniformly on every compact subinterval of I , the sum $s(x) = \sum f_n(x)$ is continuously differentiable, and*

$$s'(x) = \sum f'_n(x).$$

Proof. Fix a compact interval $[a, b] \subset I$ containing x_0 . By the fundamental theorem of calculus,

$$f_n(x) = f_n(x_0) + \int_{x_0}^x f'_n(t) dt, \quad x \in [a, b].$$

The uniform convergence of $\sum f'_n$ on $[a, b]$ together with the theorem on term-by-term integration yields that the series

$$\sum (f_n(x) - f_n(x_0))$$

converges uniformly on $[a, b]$. Since $\sum f_n(x_0)$ converges by hypothesis, we conclude that $\sum f_n$ converges uniformly on $[a, b]$.

Let $s(x) = \sum f_n(x)$. Then

$$s(x) = s(x_0) + \int_{x_0}^x \sum f'_n(t) dt.$$

The integrand on the right is continuous (uniform limit of continuous functions), so s is continuously differentiable and $s' = \sum f'_n$. $\square$

1.6 Power series

The preceding results apply particularly cleanly to power series.

Theorem 1.6.1. *Let $\sum_{n=0}^{\infty} a_n(x - c)^n$ be a power series with radius of convergence $R > 0$. Then the series converges uniformly on every compact interval contained in the open interval of convergence $(c - R, c + R)$. Moreover, the sum is infinitely differentiable inside the interval of convergence, and the derivatives may be obtained by term-by-term differentiation (the resulting series having the same radius of convergence).*

The proof follows the standard pattern: on any interval $|x - c| \leq r < R$ one has $|a_n(x - c)^n| \leq |a_n|r^n$, and $\sum |a_n|r^n$ converges, so the Weierstrass M -test applies. Term-by-term differentiation is then justified by the theorem of the previous section.

1.7 The Weierstrass approximation theorem

We conclude this chapter with one of the deepest classical applications of uniform convergence.

Theorem (Weierstrass). *Every continuous real-valued function on a compact interval $[a, b]$ is the uniform limit of a sequence of polynomials.*

There are many proofs. One of the most elegant proceeds via the Bernstein polynomials. For $f \in C[0, 1]$ one defines

$$B_n(f; x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1 - x)^{n-k}.$$

A direct (though slightly technical) estimate using the binomial theorem and the continuity of f shows that $\|B_n(f) - f\|_{[0,1]} \rightarrow 0$. The general case of an arbitrary interval $[a, b]$ follows by a linear change of variable.

An alternative proof uses the Stone–Weierstrass theorem in the special case of the algebra of polynomials, but that argument belongs to a more abstract setting and is omitted here.

Remark 1.7.1. The Weierstrass theorem is remarkable because it asserts that the seemingly rigid class of polynomials is dense, with respect to the uniform norm, in the much larger space of continuous functions. This density is the starting point for many constructive approximation procedures and for the modern theory of Banach algebras.

Notes and references for Chapter 1

The material of this chapter is classical. The systematic development of uniform convergence goes back to Weierstrass and to the Italian school of the late nineteenth century. Modern expositions can be found in:

- W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw-Hill, 1976, Chapter 7.
- T. M. Apostol, *Mathematical Analysis*, 2nd ed., Addison-Wesley, 1974, Chapter 9.
- V. A. Zorich, *Mathematical Analysis I*, Springer, 2004, Chapter 16.
- B. R. Gelbaum and J. M. H. Olmsted, *Counterexamples in Analysis*, Dover, 2003 (for the sharpness of the hypotheses).

The proof of the Weierstrass approximation theorem via Bernstein polynomials is due to S. N. Bernstein (1912). A detailed elementary account appears in many analysis texts; see, for instance, Rudin or Apostol.

End of Chapter 1

Chapter 2

The Riemann Integral: Deeper Aspects

2.1 Improper integrals of the first kind

Let f be defined on a half-open interval $[a, +\infty)$. We say that the improper integral

$$\int_a^{+\infty} f(x) dx$$

converges if the limit

$$\lim_{b \rightarrow +\infty} \int_a^b f(x) dx$$

exists and is finite. In that case the value of the improper integral is defined to be this limit. Otherwise the integral is said to *diverge*.

An analogous definition applies to intervals of the form $(-\infty, b]$ and to the two-sided improper integral over $\mathbb{R}$.

Proposition 2.1.1 (Comparison test). *Suppose $0 \leq f(x) \leq g(x)$ for all sufficiently large x . If $\int_a^{+\infty} g$ converges, then $\int_a^{+\infty} f$ converges. If $\int_a^{+\infty} f$ diverges, then $\int_a^{+\infty} g$ diverges.*

The proof is immediate from the monotonicity of the integral and the definition of the limit.

Example 2.1.2. The integral $\int_1^{+\infty} x^{-p} dx$ converges if and only if $p > 1$. Indeed,

$$\int_1^b x^{-p} dx = \begin{cases} \frac{b^{1-p} - 1}{1-p} & \text{if } p \neq 1, \\ \log b & \text{if } p = 1. \end{cases}$$

Passing to the limit $b \rightarrow +\infty$ yields the claim.

Theorem 2.1.3 (Absolute convergence implies convergence). *If $\int_a^{+\infty} |f(x)| dx$ converges, then $\int_a^{+\infty} f(x) dx$ converges.*

Proof. Write $f = f^+ - f^-$, where $f^+ = \max\{f, 0\}$ and $f^- = \max\{-f, 0\}$. Both f^+ and f^- are non-negative and dominated by $|f|$. Hence both improper integrals of f^+ and f^- converge by the comparison test, and therefore so does their difference. $\square$

The converse is false: the integral $\int_1^{+\infty} \frac{\sin x}{x} dx$ converges (by Dirichlet's test, see below), yet $\int_1^{+\infty} \left| \frac{\sin x}{x} \right| dx$ diverges.

2.2 Dirichlet's test for improper integrals

Theorem 2.2.1 (Dirichlet). *Let f and g be defined on $[a, +\infty)$. Suppose that*

- (i) *the function $F(b) = \int_a^b f(x) dx$ remains bounded as $b \rightarrow +\infty$;*
- (ii) *g is monotone and $g(x) \rightarrow 0$ as $x \rightarrow +\infty$.*

Then $\int_a^{+\infty} f(x)g(x) dx$ converges.

Proof. Integrate by parts (in the Stieltjes sense, or via the ordinary formula after approximating g by smooth functions if necessary). Set

$$F(x) = \int_a^x f(t) dt.$$

Then

$$\int_a^b fg = F(b)g(b) - F(a)g(a) - \int_a^b F(x) dg(x).$$

The boundary term $F(b)g(b)$ tends to zero because F is bounded and $g \rightarrow 0$. The integral on the right converges absolutely (hence converges) because $|F|$ is bounded and g is of bounded variation on $[a, +\infty)$ (being monotone and convergent). $\square$

Corollary 2.2.2. $\int_1^{+\infty} \frac{\sin x}{x} dx$ converges.

2.3 Improper integrals of the second kind

Now suppose f is defined on a half-open interval $(a, b]$ and may become unbounded near a . The improper integral

$$\int_a^b f(x) dx := \lim_{\varepsilon \rightarrow 0^+} \int_{a+\varepsilon}^b f(x) dx$$

is said to converge when the limit exists and is finite.

The most important example is

$$\int_0^1 x^{-p} dx,$$

which converges if and only if $p < 1$.

Many of the comparison and absolute-convergence tests carry over with only minor changes of language.

2.4 Integrals depending on a parameter

Let $f(x, t)$ be defined for $x \in [a, b]$ and $t \in T$, where $T \subset \mathbb{R}$ is an interval. Assume that for each fixed t the function $x \mapsto f(x, t)$ is Riemann-integrable on $[a, b]$. One may then form the function

$$F(t) = \int_a^b f(x, t) dx.$$

The central questions are: under what conditions is F continuous? Differentiable? Can one differentiate under the integral sign?

Theorem 2.4.1 (Continuity under the integral sign). *Suppose f is continuous on the compact rectangle $[a, b] \times [c, d]$. Then F is continuous on $[c, d]$.*

Proof. Uniform continuity of f on the compact set $[a, b] \times [c, d]$ implies that for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|t - t'| < \delta \implies |f(x, t) - f(x, t')| < \frac{\varepsilon}{b - a}$$

for all $x \in [a, b]$. Integrating with respect to x yields $|F(t) - F(t')| < \varepsilon$. $\square$

A more refined result, useful when the domain is unbounded, relies on uniform convergence or dominated convergence in the Riemann setting.

Theorem 2.4.2 (Differentiation under the integral sign). *Suppose f and $\partial f / \partial t$ are continuous on the rectangle $[a, b] \times (c, d)$. Then F is continuously differentiable on (c, d) and*

$$F'(t) = \int_a^b \frac{\partial f}{\partial t}(x, t) dx.$$

Proof. Fix $t_0 \in (c, d)$ and consider the difference quotient

$$\frac{F(t) - F(t_0)}{t - t_0} = \int_a^b \frac{f(x, t) - f(x, t_0)}{t - t_0} dx.$$

By the mean-value theorem, for each x there exists $\theta = \theta(x, t)$ between t and t_0 such that

$$\frac{f(x, t) - f(x, t_0)}{t - t_0} = \frac{\partial f}{\partial t}(x, \theta).$$

As $t \rightarrow t_0$, the integrand converges pointwise to $\partial f / \partial t(x, t_0)$. Moreover, by continuity of the partial derivative on a compact set, the convergence is uniform in x . Therefore one may pass to the limit under the integral sign (Chapter 1) and obtain the asserted formula. Continuity of F' follows from the previous theorem applied to $\partial f / \partial t$. $\square$

Remark 2.4.3. The continuity assumption on $\partial f / \partial t$ can be relaxed, but the version stated above is sufficient for most applications that arise in a second-year course and already illustrates the essential mechanism: uniform control of the difference quotients.

2.5 The Leibniz rule for variable limits

A useful companion result concerns integrals whose limits of integration depend on the parameter.

Theorem 2.5.1 (Leibniz rule). *Let f be continuous on a region containing the graph of two continuously differentiable functions $\alpha(t)$ and $\beta(t)$, and suppose $\partial f / \partial t$ exists and is continuous. Then*

$$\frac{d}{dt} \int_{\alpha(t)}^{\beta(t)} f(x, t) dx = f(\beta(t), t) \beta'(t) - f(\alpha(t), t) \alpha'(t) + \int_{\alpha(t)}^{\beta(t)} \frac{\partial f}{\partial t}(x, t) dx.$$

The proof consists in writing the integral as $F(\beta(t), t) - F(\alpha(t), t)$, where

$$F(u, t) = \int_a^u f(x, t) dx$$

(with a fixed lower limit a), differentiating by the chain rule, and applying the fundamental theorem of calculus together with the differentiation-under-the-integral theorem already proved.

2.6 An application: the Gaussian integral via differentiation under the sign

As an illustration of the power of the preceding theorems we sketch a classical evaluation.

Consider the function

$$I(t) = \int_0^{+\infty} e^{-tx^2} dx, \quad t > 0.$$

(The integral converges by comparison with a geometric series after the substitution $u = x\sqrt{t}$.) Differentiating under the integral sign (justified by a dominated-convergence argument on compact intervals of t and passage to the limit) yields

$$I'(t) = - \int_0^{+\infty} x^2 e^{-tx^2} dx.$$

A further integration by parts or a differential equation argument leads to the relation $I'(t) = -\frac{1}{2}t^{-1}I(t)$. Solving the differential equation and using the known value $I(1) = \sqrt{\pi}/2$ produces the famous formula

$$\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}.$$

(The rigorous justification of differentiation under the improper integral requires a little extra care with tails, but the necessary estimates are elementary.)

2.7 Final remarks on the Riemann theory

The results of this chapter show that the classical Riemann integral, despite its well-known limitations with respect to pointwise limits, remains a powerful and flexible tool once uniform convergence or suitable domination is available. Many of the theorems that are nowadays stated in the language of the Lebesgue integral admit perfectly rigorous and elementary proofs within the Riemann framework, provided one is willing to impose slightly stronger regularity assumptions (continuity of the integrand or of its partial derivatives, monotonicity, etc.).

For a second-year student the material presented here already opens the door to a large number of concrete analytic investigations: evaluation of special integrals, study of parametric dependence in physical models, and the first steps toward Fourier analysis.

Notes and references for Chapter 2

Standard references for the deeper theory of the Riemann integral and parameter-dependent integrals include:

- W. Rudin, *Principles of Mathematical Analysis*, Chapters 6 and 7.
- T. M. Apostol, *Mathematical Analysis*, Chapters 7 and 10.
- G. M. Fikhtengol'ts, *The Fundamentals of Mathematical Analysis*, Vol. 2 (especially the chapters on improper integrals and integrals depending on a parameter).
- V. A. Zorich, *Mathematical Analysis II*, Springer, 2004.

Dirichlet's test for integrals is classical; a clear modern exposition appears in Apostol. The differentiation-under-the-integral theorem in the form given here is essentially the one found in most rigorous second-year courses.

End of Chapter 2

Chapter 3

Differential Calculus of Functions of Several Variables

3.1 Preliminaries: the Euclidean space $\mathbb{R}^n$

We work throughout in the Euclidean space $\mathbb{R}^n$ equipped with the standard inner product

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

and the induced norm $\|x\| = \sqrt{\langle x, x \rangle}$. Open and closed balls, open sets, and the notions of limit and continuity for vector-valued functions $f : U \rightarrow \mathbb{R}^m$ (where $U \subset \mathbb{R}^n$ is open) are defined in the usual way via the norm. In particular, a function f is continuous at a point $a \in U$ if and only if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

We shall be concerned mainly with real-valued functions ($m = 1$), although most of the statements extend immediately to the vector-valued case by considering components.

3.2 Partial derivatives and directional derivatives

Let $f : U \rightarrow \mathbb{R}$ be defined on an open set $U \subset \mathbb{R}^n$ and let $a \in U$. The *partial derivative* of f with respect to the i -th variable at a is

$$\partial_i f(a) = \frac{\partial f}{\partial x_i}(a) := \lim_{h \rightarrow 0} \frac{f(a + h e_i) - f(a)}{h},$$

provided the limit exists (here e_i denotes the i -th standard basis vector).

More generally, if $v \in \mathbb{R}^n$ is any vector, the *directional derivative* of f at a in the direction v is

$$D_v f(a) := \lim_{t \rightarrow 0} \frac{f(a + tv) - f(a)}{t},$$

again provided the limit exists. When $v = e_i$ one recovers the partial derivative $\partial_i f(a)$.

Existence of all directional derivatives does *not* imply continuity, nor even the existence of a differential. A much stronger notion is required.

3.3 Differentiability

Definition 3.3.1. A function $f : U \rightarrow \mathbb{R}$ is said to be *differentiable* (or *Fréchet differentiable*) at a point $a \in U$ if there exists a linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - L(h)}{\|h\|} = 0.$$

The linear map L is uniquely determined when it exists; it is called the *differential* (or *total derivative*) of f at a and is denoted by $df(a)$ or $Df(a)$.

In coordinates, if f is differentiable at a , then necessarily all partial derivatives exist and

$$df(a)(h) = \sum_{i=1}^n \partial_i f(a) h_i = \langle \nabla f(a), h \rangle,$$

where $\nabla f(a) = (\partial_1 f(a), \dots, \partial_n f(a))$ is the *gradient* of f at a .

Theorem 3.3.2 (Sufficient condition for differentiability). *If all partial derivatives $\partial_i f$ exist in a neighbourhood of a and are continuous at a , then f is differentiable at a .*

Proof. We treat the case $n = 2$ for notational simplicity; the general case is identical in spirit. Write

$$\begin{aligned} f(a_1 + h_1, a_2 + h_2) - f(a_1, a_2) &= [f(a_1 + h_1, a_2 + h_2) - f(a_1, a_2 + h_2)] \\ &\quad + [f(a_1, a_2 + h_2) - f(a_1, a_2)]. \end{aligned}$$

By the mean-value theorem there exist $\theta_1, \theta_2 \in (0, 1)$ such that the first difference equals $\partial_1 f(a_1 + \theta_1 h_1, a_2 + h_2) h_1$ and the second equals $\partial_2 f(a_1, a_2 + \theta_2 h_2) h_2$. Consequently

$$\begin{aligned} f(a+h) - f(a) - \partial_1 f(a) h_1 - \partial_2 f(a) h_2 \\ &= (\partial_1 f(a_1 + \theta_1 h_1, a_2 + h_2) - \partial_1 f(a)) h_1 \\ &\quad + (\partial_2 f(a_1, a_2 + \theta_2 h_2) - \partial_2 f(a)) h_2. \end{aligned}$$

Dividing by $\|h\|$ and using the continuity of the partial derivatives at a , one sees that the right-hand side tends to zero as $h \rightarrow 0$. □

Remark 3.3.3. Continuity of the partial derivatives is sufficient but not necessary for differentiability. There exist functions that are differentiable everywhere yet whose partial derivatives fail to be continuous.

3.4 The chain rule

Theorem 3.4.1 (Chain rule). *Let $g : V \rightarrow U$ be differentiable at $b \in V \subset \mathbb{R}^m$ and let $f : U \rightarrow \mathbb{R}$ be differentiable at $a = g(b)$. Then the composition $f \circ g$ is differentiable at b and*

$$d(f \circ g)(b) = df(a) \circ dg(b).$$

In particular, if $m = 1$ and g is a path, one recovers the familiar formula

$$\frac{d}{dt}(f \circ g)(t) = \langle \nabla f(g(t)), g'(t) \rangle.$$

The proof consists in writing the increment of the composition and using the definitions of differentiability of f and of g ; the details are standard and may be found in any rigorous text.

3.5 Higher-order derivatives and Taylor's formula

If the first partial derivatives of f themselves admit partial derivatives, one obtains the second-order partial derivatives $\partial_i \partial_j f$, and so on. When all partial derivatives of order k exist and are continuous, one says that f is of class C^k .

Theorem 3.5.1 (Equality of mixed partials). *If $\partial_i \partial_j f$ and $\partial_j \partial_i f$ exist in a neighbourhood of a and are continuous at a , then*

$$\partial_i \partial_j f(a) = \partial_j \partial_i f(a).$$

(The classical proof proceeds by examining the second difference quotient and applying the mean-value theorem twice.)

Theorem 3.5.2 (Taylor's formula with Lagrange remainder). *Let f be of class C^{k+1} in a convex neighbourhood of a point $a \in \mathbb{R}^n$. Then for every h sufficiently small there exists $\theta \in (0, 1)$ such that*

$$f(a + h) = \sum_{|\alpha| \leq k} \frac{1}{\alpha!} \partial^\alpha f(a) h^\alpha + \sum_{|\alpha| = k+1} \frac{1}{\alpha!} \partial^\alpha f(a + \theta h) h^\alpha,$$

where the sums run over multi-indices and the usual multi-index notation is employed.

In the special case $k = 1$ one obtains the first-order expansion

$$f(a + h) = f(a) + \langle \nabla f(a), h \rangle + \frac{1}{2} \langle H_f(a + \theta h) h, h \rangle,$$

where H_f denotes the Hessian matrix of second partial derivatives.

3.6 Local extrema and the second-derivative test

Definition 3.6.1. A point $a \in U$ is a *local minimum* (respectively, *local maximum*) of f if there exists a neighbourhood V of a such that $f(x) \geq f(a)$ (respectively, $f(x) \leq f(a)$) for all $x \in V$.

Theorem 3.6.2 (First-order necessary condition). *If f is differentiable at an interior local extremum a , then $\nabla f(a) = 0$.*

Theorem 3.6.3 (Second-order sufficient condition). *Suppose f is of class C^2 near a and $\nabla f(a) = 0$. Let $H_f(a)$ be the Hessian matrix at a .*

- *If $H_f(a)$ is positive definite, then a is a strict local minimum.*
- *If $H_f(a)$ is negative definite, then a is a strict local maximum.*

- If $H_f(a)$ is indefinite, then a is a saddle point.

The proof of the sufficient condition follows at once from the second-order Taylor expansion: the quadratic form $\frac{1}{2}\langle H_f(a)h, h \rangle$ dominates the remainder for small h , and the definiteness of the Hessian determines the sign of $f(a+h) - f(a)$.

3.7 The implicit function theorem

One of the deepest and most useful results of multivariable differential calculus is the following.

Theorem 3.7.1 (Implicit function theorem). *Let $F : U \times V \rightarrow \mathbb{R}^m$ be a mapping of class C^1 , where $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ are open. Suppose $F(a, b) = 0$ and that the partial Jacobian matrix*

$$D_y F(a, b) = \left(\frac{\partial F_i}{\partial y_j}(a, b) \right)_{1 \leq i, j \leq m}$$

is invertible. Then there exist neighbourhoods U_0 of a and V_0 of b , and a unique mapping $g : U_0 \rightarrow V_0$ of class C^1 , such that

$$F(x, g(x)) = 0 \quad \text{for all } x \in U_0,$$

and $g(a) = b$. Moreover,

$$Dg(x) = -\left(D_y F(x, g(x))\right)^{-1} D_x F(x, g(x)).$$

The proof proceeds by reducing the problem to the inverse-function theorem (or by a direct contraction-mapping argument). For the purposes of a second-year course it is sufficient to understand the statement thoroughly and to be able to compute the derivatives of the implicitly defined function via the formula above.

Corollary 3.7.2 (Inverse function theorem). *Let $f : U \rightarrow \mathbb{R}^n$ be of class C^1 and suppose that the Jacobian matrix $Df(a)$ is invertible at a point $a \in U$. Then f is locally invertible near a , the local inverse is of class C^1 , and*

$$D(f^{-1})(f(a)) = (Df(a))^{-1}.$$

3.8 Constrained extrema: the method of Lagrange multipliers

Theorem 3.8.1 (Lagrange multipliers). *Let $f, g_1, \dots, g_m : U \rightarrow \mathbb{R}$ be of class C^1 on an open set $U \subset \mathbb{R}^n$, with $m < n$. Suppose $a \in U$ is a local extremum of f subject to the constraints $g_1(x) = \dots = g_m(x) = 0$, and that the gradients $\nabla g_1(a), \dots, \nabla g_m(a)$ are linearly independent. Then there exist real numbers $\lambda_1, \dots, \lambda_m$ (the Lagrange multipliers) such that*

$$\nabla f(a) = \sum_{k=1}^m \lambda_k \nabla g_k(a).$$

Geometrically the condition asserts that the gradient of the objective function is a linear combination of the gradients of the constraint functions, i.e., that $\nabla f(a)$ is orthogonal to the tangent space of the constraint manifold at a .

The proof may be obtained by applying the implicit-function theorem to solve the constraints locally and then imposing the first-order necessary condition on the resulting unconstrained function of $n - m$ variables.

Notes and references for Chapter 3

The differential calculus of several variables is treated at a high level of rigour in:

- W. Rudin, *Principles of Mathematical Analysis*, Chapter 9.
- T. M. Apostol, *Mathematical Analysis*, Chapter 12.
- M. Spivak, *Calculus on Manifolds*, W. A. Benjamin, 1965 (especially the early chapters).
- V. A. Zorich, *Mathematical Analysis II*, Chapters 8–10.
- J. Munkres, *Analysis on Manifolds*, Addison-Wesley, 1991.

The implicit-function theorem and the method of Lagrange multipliers appear in every serious multivariable analysis course; the versions given above are essentially those of Rudin and Apostol. For a more geometric perspective one may consult Spivak or Munkres.

End of Chapter 3

Chapter 4

Multiple Integrals

4.1 Definition of the Riemann integral over a rectangle

Let $R = [a_1, b_1] \times \cdots \times [a_n, b_n] \subset \mathbb{R}^n$ be a closed rectangle and let $f : R \rightarrow \mathbb{R}$ be a bounded function. A partition P of R is given by partitions of each of the intervals $[a_i, b_i]$. The corresponding subrectangles are denoted R_j , and one forms the upper and lower Darboux sums

$$U(f, P) = \sum_j M_j \operatorname{vol}(R_j), \quad L(f, P) = \sum_j m_j \operatorname{vol}(R_j),$$

where $M_j = \sup_{R_j} f$ and $m_j = \inf_{R_j} f$. The function f is said to be *Riemann integrable* over R if

$$\inf_P U(f, P) = \sup_P L(f, P).$$

The common value is denoted $\int_R f$.

As in one variable, a bounded function on a rectangle is Riemann integrable if and only if its set of points of discontinuity has measure zero (Lebesgue's criterion). In particular every continuous function on a compact rectangle is integrable.

4.2 Fubini's theorem for continuous functions

The most important practical tool for evaluating multiple integrals is the reduction to iterated integrals.

Theorem 4.2.1 (Fubini). *Let f be continuous on the rectangle $R = [a, b] \times [c, d] \subset \mathbb{R}^2$. Then*

$$\int_R f(x, y) \, dx \, dy = \int_a^b \left(\int_c^d f(x, y) \, dy \right) dx = \int_c^d \left(\int_a^b f(x, y) \, dx \right) dy.$$

Proof. The inner integral $F(x) = \int_c^d f(x, y) \, dy$ is continuous in x by the theorem on continuity under the integral sign (Chapter 2). The resulting ordinary integral $\int_a^b F(x) \, dx$ therefore exists. To identify it with the double integral one compares upper and lower sums: for any partition of the rectangle the upper (respectively lower) double sum dominates (respectively is dominated by) the corresponding iterated upper (lower) sum. Taking the appropriate limits yields equality. $\square$

The same statement holds, with only notational changes, for continuous functions on rectangles in $\mathbb{R}^n$ and for the corresponding n -fold iterated integrals.

Remark 4.2.2. If f is merely Riemann integrable (not necessarily continuous), Fubini's theorem still holds provided one interprets the inner integrals carefully (they exist for almost every fixed value of the remaining variables). The continuous case already covers the great majority of concrete computations that arise in a second-year course.

4.3 Integrals over elementary regions

Many domains that appear in applications are not rectangles. An *elementary region* in the plane (of type I) is a set of the form

$$D = \{(x, y) : a \leq x \leq b, \varphi(x) \leq y \leq \psi(x)\},$$

where φ and ψ are continuous. One defines

$$\int_D f := \int_a^b \left(\int_{\varphi(x)}^{\psi(x)} f(x, y) dy \right) dx$$

whenever the iterated integral exists. An analogous definition holds for regions of type II (described by functions of y).

If f is continuous on a compact elementary region D , the integral is well-defined and independent of the particular description of D . Moreover, if D is contained in a rectangle R and one extends f by zero outside D , the resulting function is Riemann integrable on R and the two notions of integral coincide.

4.4 Change of variables

The change-of-variables formula is the higher-dimensional analogue of the substitution rule.

Theorem 4.4.1 (Change of variables). *Let $U \subset \mathbb{R}^n$ be open and let $\Phi : U \rightarrow \mathbb{R}^n$ be a C^1 -diffeomorphism onto its image (i.e., Φ is bijective, of class C^1 , and $D\Phi(x)$ is invertible for every $x \in U$). Let f be continuous and compactly supported in $\Phi(U)$. Then*

$$\int_{\Phi(U)} f(y) dy = \int_U (f \circ \Phi)(x) |\det D\Phi(x)| dx.$$

In dimension two the absolute value of the Jacobian determinant $|\det D\Phi|$ measures the local distortion of area. The formula is first proved for linear maps (where it reduces to the elementary fact that the volume of a parallelepiped is the absolute value of the determinant of its spanning vectors), then for C^1 maps by approximation, and finally extended to the general case by a partition-of-unity argument or by reducing to the linear case via the inverse-function theorem.

Example 4.4.2 (Polar coordinates). The mapping

$$\Phi(r, \theta) = (r \cos \theta, r \sin \theta), \quad r > 0, \theta \in (0, 2\pi),$$

has Jacobian determinant equal to r . Consequently

$$\iint_D f(x, y) \, dx \, dy = \iint_{\Phi^{-1}(D)} f(r \cos \theta, r \sin \theta) r \, dr \, d\theta$$

for suitable regions D . This transformation is indispensable for integrals involving radial symmetry (e.g., the Gaussian integral already encountered in Chapter 2).

4.5 Improper multiple integrals

As in one variable, one may define improper integrals over unbounded regions or over regions where the integrand becomes singular by taking limits of integrals over exhausting compact subregions. Absolute convergence again implies ordinary convergence, and comparison tests remain valid.

A typical example is the integral

$$\iint_{\mathbb{R}^2} e^{-(x^2+y^2)} \, dx \, dy = \pi,$$

which is most easily evaluated in polar coordinates and yields, as a corollary, the value of the one-dimensional Gaussian integral.

4.6 Line integrals and Green's theorem (brief outlook)

Although a full treatment of differential forms lies beyond the scope of a second-year course, it is worth recording the classical Green theorem, which already illustrates the deep connection between integration and differentiation in several variables.

Theorem 4.6.1 (Green). *Let P and Q be of class C^1 on an open set containing a compact elementary region D whose boundary ∂D consists of finitely many piecewise C^1 curves, oriented positively with respect to D . Then*

$$\int_{\partial D} P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dx \, dy.$$

Green's theorem is the two-dimensional prototype of the general Stokes theorem. Its proof for elementary regions follows by writing the double integral as an iterated integral and applying the fundamental theorem of calculus in each variable separately.

Notes and references for Chapter 4

The theory of multiple integrals is developed rigorously in:

- W. Rudin, *Principles of Mathematical Analysis*, Chapter 10 (especially the treatment of the change-of-variables formula).
- T. M. Apostol, *Mathematical Analysis*, Chapter 14.
- M. Spivak, *Calculus on Manifolds*, Chapters 3–5.
- V. A. Zorich, *Mathematical Analysis II*, Chapters 11–13.
- J. Munkres, *Analysis on Manifolds*, Chapters 4–7.

Fubini's theorem for continuous functions admits a completely elementary proof via Darboux sums, as indicated above. The general change-of-variables theorem is considerably deeper; the version stated here is sufficient for all standard applications that appear in the undergraduate curriculum.

End of Chapter 4

Chapter 5

Fourier Series

5.1 Trigonometric polynomials and orthogonal systems

A *trigonometric polynomial* of order N is a function of the form

$$T(x) = \frac{a_0}{2} + \sum_{k=1}^N (a_k \cos kx + b_k \sin kx).$$

The system

$$\{1, \cos x, \sin x, \cos 2x, \sin 2x, \dots\}$$

is orthogonal on the interval $[-\pi, \pi]$ with respect to the inner product

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(x)g(x) dx.$$

More precisely,

$$\int_{-\pi}^{\pi} \cos mx \cos nx dx = \begin{cases} 0 & m \neq n, \\ \pi & m = n \geq 1, \\ 2\pi & m = n = 0, \end{cases}$$

and analogous formulae hold for the sine functions and for mixed products (which vanish).

Consequently, if a function f admits an expansion

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx)$$

that converges reasonably well, the coefficients are necessarily given by the Euler–Fourier formulae

$$\begin{aligned} a_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx dx, & k \geq 0, \\ b_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin kx dx, & k \geq 1. \end{aligned}$$

5.2 The Fourier series of an integrable function

Definition 5.2.1. Let f be Riemann integrable on $[-\pi, \pi]$ (or, more generally, absolutely integrable). The *Fourier series* of f is the trigonometric series

$$\frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx)$$

whose coefficients are given by the Euler–Fourier formulae above. We write

$$f \sim \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx).$$

The symbol $\sim$ emphasises that the series is formally associated with f ; convergence questions are separate and non-trivial.

Remark 5.2.2. If f is defined on an arbitrary interval $[a, a + 2\pi]$, one reduces to the standard interval by a linear change of variable. Even and odd extensions yield pure cosine or pure sine series on $[0, \pi]$.

5.3 Partial sums and the Dirichlet kernel

The partial sum of the Fourier series is

$$S_N(f; x) = \frac{a_0}{2} + \sum_{k=1}^N (a_k \cos kx + b_k \sin kx).$$

A direct computation using the orthogonality relations produces the integral representation

$$S_N(f; x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x-t) D_N(t) dt,$$

where

$$D_N(t) = \frac{1}{2} + \sum_{k=1}^N \cos kt = \frac{\sin((N + \frac{1}{2})t)}{2 \sin(t/2)}$$

is the *Dirichlet kernel*. The identity

$$\frac{1}{\pi} \int_{-\pi}^{\pi} D_N(t) dt = 1$$

shows that $S_N(f; x)$ is a weighted average of the values of f .

5.4 Pointwise convergence

Theorem 5.4.1 (Dirichlet–Jordan test). *Suppose f is of bounded variation on $[-\pi, \pi]$ (in particular, if f is piecewise monotone or piecewise continuously differentiable). Then at every point x*

the partial sums $S_N(f; x)$ converge to

$$\frac{f(x+) + f(x-)}{2},$$

where $f(x\pm)$ denote the one-sided limits. In particular, at every point of continuity one has $S_N(f; x) \rightarrow f(x)$.

The proof relies on the Riemann–Lebesgue lemma (the Fourier coefficients of an integrable function tend to zero) together with a careful analysis of the Dirichlet kernel near the origin and away from it. For functions of class C^1 a simpler integration-by-parts argument already yields uniform convergence of the differentiated series and therefore uniform convergence of the original series.

Theorem 5.4.2 (Uniform convergence for smooth functions). *If f is 2π -periodic and of class C^1 , then the Fourier series of f converges to f uniformly and absolutely on $\mathbb{R}$.*

Sketch. Integration by parts shows that $a_k = O(1/k)$ and $b_k = O(1/k)$. A second integration by parts (using f'') improves the decay to $O(1/k^2)$, which is enough for absolute and uniform convergence by the Weierstrass M -test. $\square$

5.5 Parseval's identity and mean-square convergence

Even when pointwise convergence fails, the Fourier series still converges to f in a weaker but very useful sense.

Theorem 5.5.1 (Parseval). *If f is Riemann integrable on $[-\pi, \pi]$, then*

$$\frac{1}{\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx = \frac{a_0^2}{2} + \sum_{k=1}^{\infty} (a_k^2 + b_k^2).$$

In particular the partial sums $S_N(f)$ converge to f in the L^2 norm:

$$\int_{-\pi}^{\pi} |f(x) - S_N(f; x)|^2 dx \rightarrow 0.$$

The identity follows from the orthogonality of the trigonometric system once one knows that trigonometric polynomials are dense in the mean-square sense among continuous (or Riemann-integrable) functions — a consequence of the Weierstrass approximation theorem (Chapter 1) applied to the continuous periodic case, or of Fejér's theorem on Cesàro means.

5.6 Fejér's theorem

A remarkable positive result that requires no extra smoothness is the following.

Theorem 5.6.1 (Fejér). *Let $\sigma_N(f)$ denote the average of the first $N + 1$ partial sums of the Fourier series of a continuous 2π -periodic function f . Then $\sigma_N(f)$ converges to f uniformly on $\mathbb{R}$.*

The proof uses the non-negativity and approximate-identity properties of the Fejér kernel

$$K_N(t) = \frac{1}{N+1} \sum_{k=0}^N D_k(t) = \frac{1}{2(N+1)} \left(\frac{\sin((N+1)t/2)}{\sin(t/2)} \right)^2.$$

This theorem supplies an independent proof that trigonometric polynomials are dense in the uniform norm among continuous periodic functions, and therefore yields a new approach to the classical Weierstrass approximation theorem on the circle.

5.7 Applications and further remarks

Fourier series constitute one of the most powerful tools of classical analysis. They permit the explicit solution of the heat equation and the wave equation on an interval (by separation of variables), the analysis of periodic phenomena in physics and engineering, and the construction of continuous nowhere-differentiable functions (Weierstrass). The deeper theory — absolute convergence criteria, localization principles, Carleson's theorem on almost-everywhere convergence of Fourier series of L^2 functions — lies beyond the scope of a second-year course, yet the elementary results already presented open a wide field of concrete investigations.

Notes and references for Chapter 5

Classical and modern treatments of Fourier series include:

- W. Rudin, *Principles of Mathematical Analysis*, Chapter 8 (brief but precise).
- T. M. Apostol, *Mathematical Analysis*, Chapter 11.
- G. M. Fikhtengol'ts, *The Fundamentals of Mathematical Analysis*, Vol. 2 (detailed classical exposition).
- Y. Katznelson, *An Introduction to Harmonic Analysis*, Dover, 2004.
- E. M. Stein & R. Shakarchi, *Fourier Analysis*, Princeton, 2003 (more advanced).

Fejér's theorem and the elementary theory of the Dirichlet kernel are standard in every rigorous introduction. The version of the Dirichlet–Jordan test stated above is sufficient for all piecewise smooth functions that appear in applications.

End of Chapter 5

Chapter 6

Metric Spaces and Compactness in $\mathbb{R}^n$

6.1 Metric spaces: basic notions

Although most of the preceding chapters have been written in the concrete setting of the real line or Euclidean space, many of the arguments become clearer and more powerful when viewed in the abstract language of metric spaces. We therefore collect here the essential definitions and theorems that are needed for a rigorous second-year treatment of analysis in $\mathbb{R}^n$.

Definition 6.1.1. A *metric* on a set X is a function $d : X \times X \rightarrow [0, +\infty)$ satisfying

- (i) $d(x, y) = 0$ if and only if $x = y$,
- (ii) $d(x, y) = d(y, x)$,
- (iii) $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

The pair (X, d) is called a *metric space*.

The Euclidean space $\mathbb{R}^n$ with

$$d(x, y) = \|x - y\| = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

is the principal example. Other useful metrics on $\mathbb{R}^n$ include the ℓ^1 and ℓ^∞ metrics; all of them induce the same topology (they are equivalent).

An open ball, a closed ball, an open set, a closed set, the interior, the closure, and the boundary are defined exactly as in $\mathbb{R}$. A set $K \subset X$ is *compact* if every open cover of K admits a finite subcover. A set is *sequentially compact* if every sequence in the set has a subsequence that converges to a point of the set.

6.2 Completeness

Definition 6.2.1. A sequence $\{x_k\}$ in a metric space (X, d) is *Cauchy* if for every $\varepsilon > 0$ there exists N such that

$$m, n \geq N \implies d(x_m, x_n) < \varepsilon.$$

The space is *complete* if every Cauchy sequence converges to a point of X .

Theorem 6.2.2. $\mathbb{R}^n$ (with the Euclidean metric) is complete.

Proof. Let $\{x^{(k)}\}$ be a Cauchy sequence in $\mathbb{R}^n$. Each coordinate sequence $\{x_i^{(k)}\}_{k=1}^\infty$ is Cauchy in $\mathbb{R}$ and therefore converges. The resulting limit vector is the limit of the original sequence. $\square$

A closed subset of a complete metric space is itself complete. In particular every closed subset of $\mathbb{R}^n$ is complete.

6.3 The Heine-Borel theorem

The following characterisation of compact subsets of Euclidean space is fundamental.

Theorem 6.3.1 (Heine-Borel). *A subset $K \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Outline. Necessity is easy: a compact set is closed (its complement is open) and bounded (otherwise the open cover by balls of radius $1, 2, 3, \dots$ centred at the origin would have no finite subcover).

Sufficiency: first one shows that a closed bounded interval in $\mathbb{R}$ is compact (by the nested-interval property or by the Bolzano-Weierstrass theorem). The higher-dimensional case follows by writing a closed bounded rectangle as a product of intervals and using a standard diagonal argument, or by observing that sequential compactness is equivalent to compactness in $\mathbb{R}^n$ and applying Bolzano-Weierstrass coordinatewise. $\square$

Corollary 6.3.2 (Bolzano-Weierstrass). *Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.*

6.4 Uniform continuity on compact sets

Theorem 6.4.1. *Let $f : K \rightarrow \mathbb{R}^m$ be continuous, where $K \subset \mathbb{R}^n$ is compact. Then f is uniformly continuous on K .*

Proof. Suppose not. Then there exists $\varepsilon > 0$ such that for every $k \in \mathbb{N}$ one can find points $x_k, y_k \in K$ with

$$d(x_k, y_k) < \frac{1}{k} \quad \text{yet} \quad \|f(x_k) - f(y_k)\| \geq \varepsilon.$$

By compactness there is a subsequence $x_{k_j} \rightarrow x \in K$. The corresponding y_{k_j} also converge to the same point x . Continuity of f at x then yields a contradiction. $\square$

This theorem is the precise reason why a continuous function on a closed bounded interval attains its maximum and minimum, and why the Riemann integral of a continuous function exists.

6.5 The contraction mapping principle

We record one more abstract result that will be used in the next chapter.

Theorem 6.5.1 (Banach fixed-point theorem). *Let (X, d) be a complete metric space and let $T : X \rightarrow X$ be a contraction, i.e., there exists $\lambda \in [0, 1)$ such that*

$$d(Tx, Ty) \leq \lambda d(x, y) \quad \text{for all } x, y \in X.$$

Then T has a unique fixed point $x^* \in X$. Moreover, for every $x_0 \in X$ the iterates $x_{k+1} = Tx_k$ converge to x^* .

Proof. The sequence of iterates is easily seen to be Cauchy (by summing a geometric series). Completeness supplies a limit x^* , and continuity of T shows that x^* is a fixed point. Uniqueness follows at once from the contraction inequality. $\square$

6.6 Equivalence of norms on $\mathbb{R}^n$

All norms on the finite-dimensional space $\mathbb{R}^n$ are equivalent: if $\|\cdot\|_a$ and $\|\cdot\|_b$ are any two norms, there exist positive constants c, C such that

$$c\|x\|_a \leq \|x\|_b \leq C\|x\|_a \quad \text{for all } x \in \mathbb{R}^n.$$

In particular they induce the same topology, the same convergent sequences, and the same continuous functions. This fact justifies the free use of whichever norm is most convenient in a given argument.

Notes and references for Chapter 6

The material of this chapter is standard in any rigorous treatment of multivariable analysis:

- W. Rudin, *Principles of Mathematical Analysis*, Chapter 2 (metric spaces) and Chapter 4 (continuity).
- T. M. Apostol, *Mathematical Analysis*, Chapter 4.
- V. A. Zorich, *Mathematical Analysis II*, Chapter 9.
- J. Munkres, *Topology*, Prentice-Hall, 2000 (for a more topological perspective).

The Heine-Borel theorem and the uniform continuity theorem on compact sets are among the most frequently used tools in the whole of real analysis. The contraction mapping principle will be applied in the next chapter to the existence and uniqueness theory of ordinary differential equations.

End of Chapter 6

Chapter 7

Ordinary Differential Equations: Existence and Uniqueness

7.1 The initial-value problem

We consider the first-order system

$$\mathbf{y}' = \mathbf{f}(x, \mathbf{y}), \quad \mathbf{y}(x_0) = \mathbf{y}_0, \quad (7.1)$$

where $\mathbf{f} : D \rightarrow \mathbb{R}^n$ is defined on an open set $D \subset \mathbb{R} \times \mathbb{R}^n$ and $(x_0, \mathbf{y}_0) \in D$. A solution on an interval I containing x_0 is a continuously differentiable map $\mathbf{y} : I \rightarrow \mathbb{R}^n$ such that $(x, \mathbf{y}(x)) \in D$ for all $x \in I$ and the differential equation together with the initial condition are satisfied.

The fundamental questions are existence, uniqueness, and the maximal interval of existence.

7.2 Integral equation formulation

Integrating both sides of the differential equation and using the initial condition shows that a continuously differentiable function $\mathbf{y}$ solves (7.1) if and only if it satisfies the integral equation

$$\mathbf{y}(x) = \mathbf{y}_0 + \int_{x_0}^x \mathbf{f}(t, \mathbf{y}(t)) dt.$$

This reformulation is the starting point of the Picard iteration method.

7.3 The Picard–Lindelöf theorem

Theorem 7.3.1 (Picard–Lindelöf). *Suppose $\mathbf{f}$ is continuous on an open set $D \subset \mathbb{R} \times \mathbb{R}^n$ and is locally Lipschitz continuous with respect to $\mathbf{y}$: for every compact set $K \subset D$ there exists a constant L_K such that*

$$\|\mathbf{f}(x, \mathbf{y}_1) - \mathbf{f}(x, \mathbf{y}_2)\| \leq L_K \|\mathbf{y}_1 - \mathbf{y}_2\|$$

whenever $(x, \mathbf{y}_1), (x, \mathbf{y}_2) \in K$. Then for every $(x_0, \mathbf{y}_0) \in D$ there exists $h > 0$ such that the initial-value problem (7.1) admits a unique solution on the interval $[x_0 - h, x_0 + h]$.

Proof. Choose a closed rectangle

$$R = \{(x, \mathbf{y}) : |x - x_0| \leq a, \|\mathbf{y} - \mathbf{y}_0\| \leq b\}$$

contained in D . Let

$$M = \max_R \|\mathbf{f}\|, \quad L = \text{Lipschitz constant of } \mathbf{f} \text{ on } R.$$

Set

$$h = \min \left\{ a, \frac{b}{M}, \frac{1}{2L} \right\}$$

(if $L = 0$ the third restriction is omitted). Consider the complete metric space X of continuous maps

$$\mathbf{y} : [x_0 - h, x_0 + h] \rightarrow \mathbb{R}^n$$

satisfying $\|\mathbf{y}(x) - \mathbf{y}_0\| \leq b$, equipped with the supremum metric. The Picard operator

$$(T\mathbf{y})(x) = \mathbf{y}_0 + \int_{x_0}^x \mathbf{f}(t, \mathbf{y}(t)) dt$$

maps X into itself (by the choice of h) and is a contraction with constant $\lambda = Lh \leq 1/2$. The Banach fixed-point theorem (Chapter 6) therefore yields a unique fixed point, which is the unique solution of the integral equation and hence of the initial-value problem on $[x_0 - h, x_0 + h]$. $\square$

Remark 7.3.2. Continuity of $\mathbf{f}$ alone guarantees local existence (Peano's theorem), but uniqueness may fail. The classical counter-example is $y' = \sqrt{|y|}$, $y(0) = 0$, which admits more than one solution.

7.4 Maximal interval of existence

Theorem 7.4.1. *Under the hypotheses of the Picard–Lindelöf theorem there exists a unique maximal open interval (α, β) containing x_0 on which a solution $\mathbf{y}$ of (7.1) is defined. Moreover, if $\beta < +\infty$ (respectively $\alpha > -\infty$), then the graph of $\mathbf{y}$ leaves every compact subset of D as $x \rightarrow \beta^-$ (respectively $x \rightarrow \alpha^+$).*

In other words, the only way a solution can cease to exist in finite time is by approaching the boundary of the domain D (or by blowing up).

7.5 Linear systems

When $\mathbf{f}(x, \mathbf{y}) = A(x)\mathbf{y} + \mathbf{g}(x)$ with continuous matrix-valued A and vector-valued $\mathbf{g}$, the Lipschitz condition is automatically satisfied on any strip $|x - x_0| \leq a$. Consequently linear systems with continuous coefficients possess unique solutions defined on the whole interval where the coefficients remain continuous.

7.6 Dependence on initial conditions and parameters

If $\mathbf{f}$ is of class C^1 (or merely locally Lipschitz), the unique solution depends continuously on the initial data $\mathbf{y}_0$ and on any parameters that enter continuously into $\mathbf{f}$. Under stronger smoothness assumptions the dependence is differentiable, and the derivative satisfies the associated variational

equation. These facts are the starting point for the qualitative theory of differential equations and for perturbation methods.

7.7 A simple example

Consider the scalar equation

$$y' = y^2, \quad y(0) = y_0 > 0.$$

Separation of variables yields the explicit solution

$$y(x) = \frac{y_0}{1 - y_0 x},$$

which exists on the maximal interval $(-\infty, 1/y_0)$. As $x \rightarrow (1/y_0)^-$ one has $y(x) \rightarrow +\infty$, illustrating the blow-up phenomenon described by the maximal-interval theorem.

Notes and references for Chapter 7

The existence and uniqueness theory for ordinary differential equations is treated in:

- W. Rudin, *Principles of Mathematical Analysis*, Chapter 9 (brief remarks).
- T. M. Apostol, *Mathematical Analysis*, Chapter 6 (for the one-dimensional case) and more advanced texts for systems.
- P. Hartman, *Ordinary Differential Equations*, SIAM, 2002.
- V. I. Arnold, *Ordinary Differential Equations*, Springer, 1992 (geometric viewpoint).
- E. A. Coddington & N. Levinson, *Theory of Ordinary Differential Equations*, McGraw-Hill, 1955 (classical reference).

The proof of the Picard–Lindelöf theorem given above is the standard contraction-mapping argument; it illustrates perfectly the usefulness of the abstract metric-space notions introduced in Chapter 6.

End of Chapter 7

Conclusion

The seven chapters of this monograph form a coherent path through some of the central themes of classical real analysis. Beginning with the precise notion of uniform convergence, we saw how a single analytic condition governs the preservation of continuity, the legitimacy of term-by-term integration and differentiation, and the density of polynomials among continuous functions. The subsequent study of the Riemann integral in greater depth showed that many results usually associated with Lebesgue theory already admit satisfactory elementary proofs once uniform control or suitable monotonicity is available.

The differential calculus of several variables and the theory of multiple integrals then extended the same circle of ideas to higher dimensions, culminating in the change-of-variables formula and a first glimpse of Green's theorem. Fourier series illustrated the power of orthogonal expansions and supplied, via Fejér's theorem, an independent approach to approximation by trigonometric polynomials. The language of metric spaces clarified the underlying topological structure of Euclidean space and prepared the ground for the contraction-mapping proof of the Picard–Lindelöf theorem, which closes the work with a fundamental existence and uniqueness result for ordinary differential equations.

Throughout, the emphasis has remained on complete, self-contained arguments that stay within the reach of a second-year student who has already mastered the foundations of single-variable analysis. The author hopes that the notes may serve both as a careful review of the core curriculum and as a bridge toward more advanced studies in analysis, differential equations, and the geometry of Euclidean space.

Baku, August 2026

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— End of the Monograph —